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Raman sideband cooling in a finite 3D lattice

Rb-85 spin preparation, motional dynamics and atom retention

Abstract. This simulator predicts spin preparation, axial cooling and retention of rubidium-85 in a three-dimensional optical lattice. A frozen spatial ensemble combines coherent Raman dynamics with photon recoil in all axes. Recoil beyond any local bound spectrum produces absorbing loss. The model distinguishes survivor purity from absolute target yield and estimates temperature from motional energy. After the reference 5 ms sequence, 52.1% of loaded atoms survive and 27.1% reach $|3,3; n_z=0\rangle$ while remaining bound in all axes. This paper explains the physical model, numerical methods and their limits; the results are predictions rather than measured efficiencies.

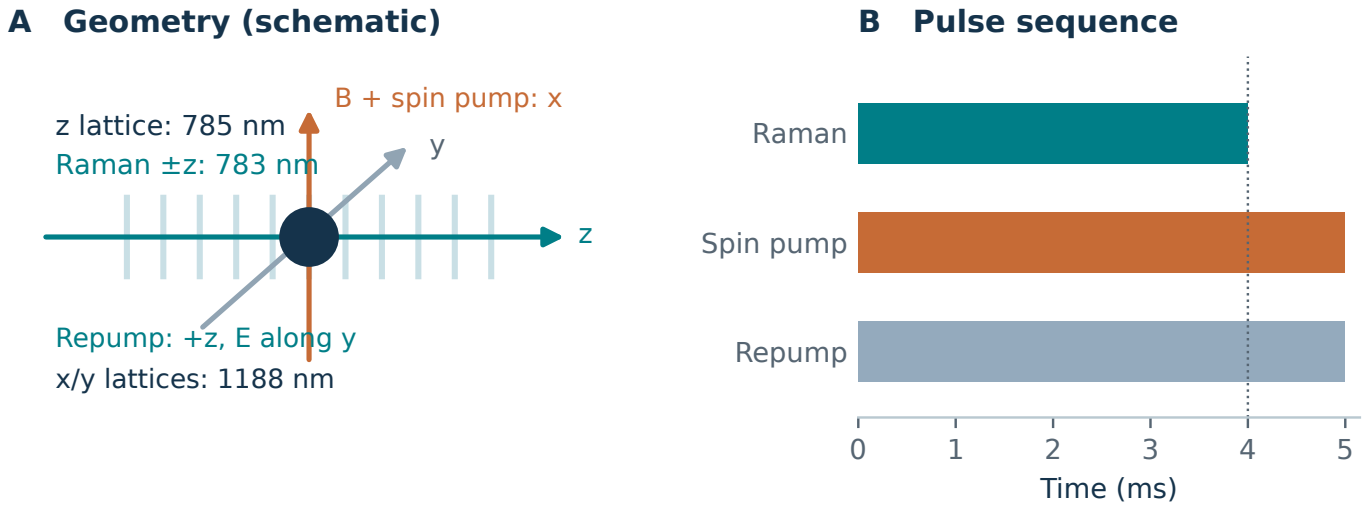


Figure 1. Geometry and programmed sequence. The bias field fixes quantization along x; Raman light addresses the axial red sideband. Both pumping beams remain on during the final millisecond. The geometry is schematic, not to scale.

1 Experiment and scope

The reference input is **examples/experiment.json**. Each Raman beam carries both equal-power tones; the counterpropagating beams have opposite helicities relative to their own directions and equal laboratory Jones vectors. The spin pump is pure σ+; the repump has linear E along y. The two transverse lattice carriers differ by 160 MHz, suppressing their mutual interference. Atomic constants and dipoles are based on Steck and ARC [1,2].

Input	Reference value
Confinement	z: 785 nm; x/y: 1188 nm; central depth 15 μK per axis
Derived bottom frequencies	69.04 kHz (z); 45.62 kHz (x/y). Depth and wavelength set the wells.
Cloud and beams	150 μm cloud density radius; 200 μm beam intensity waists (all 1/e²)
Initially loaded sample	10 μK per axis, conditioned on local bound states; uniform F = 3
Drive settings	1 G bias; 5 kHz measured carrier; spin pump 3 nW; repump 7.5 nW

2 From optical fields to state dynamics

Local wells and the retained quantum state

Gaussian beam envelopes set each sample's local depths and intensities. Each axis is a sinusoidal well continued into a flat barrier outside the site. A finite-difference eigensolve retains all levels below the barrier, including anharmonicity. Initial motional weights are proportional to $\exp[-(E_n-E_0)/(k_B T)]$ and normalized over each local bound spectrum at T = 10 µK. Loading survival is not calculated.

$$P_{s,n_x,n_y,n_z}(t), \qquad \rho^{(n_x,n_y)}_{an_z,\,bm_z}(t), \quad a,b \in \{u,d\} \tag{1}$$

Here s labels the 12 ground spin states; u = |3,3⟩ and d = |2,2⟩ form the Raman pair. At each (n_x, n_y), their full axial density matrix retains motional coherences. The u/d populations in P are its diagonal entries, not additional atoms. Other spin states carry populations. This hybrid state preserves accumulated spin-motion and inter-axis population correlations; for each sample, ΣP + L = 1, where L is absorbed loss.

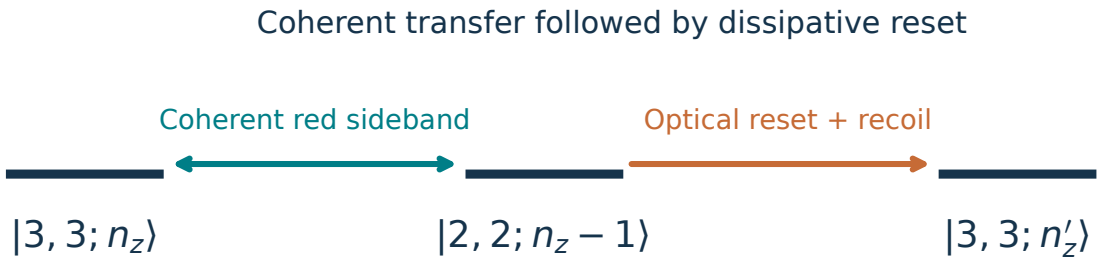


Figure 2. Intended state-transfer cycle, not an energy-level diagram. Raman transfer removes axial excitation. Reset traverses other spin states and changes n_x, n_y and n_z through recoil; n'_z need not equal n_z-1, and escape is possible.

Coherent Raman coupling and light shifts

All four beam/tone Raman pathways add with complex phases and overlaps $\langle n'|\exp(i\Delta k\cdot r)|n\rangle$. The Hamiltonian retains carrier, red- and blue-sideband couplings through $|\Delta n_z| \le 3$, so $n_z=0$ is only approximately dark. Breit-Rabi energies and diagonal local Stark shifts set the detunings. One common laboratory beat is fixed 5 kHz below the central red-sideband reference; samples are not retuned individually.

Scattering, recoil and escape

Weakly populated excited states are eliminated into pumping rates and shifts [3]. Pumping sums D2 hyperfine paths coherently; far-detuned estimates sum D1 and D2 amplitudes before squaring [2,4]. Equal-frequency absorption paths interfere, while distinct tones are averaged in spontaneous rates. For one Cartesian axis and one absorption path,

$$K_{n'n} = \left\langle \left| \langle n'| e^{i(k_{\mathrm{abs}} - k_{\mathrm{em}})x} |n\rangle \right|^2 \right\rangle_{\mathrm{em}}, \quad \ell_n = 1 - \sum_{n' \in \mathrm{bound}} K_{n'n} \tag{2}$$

The average in Eq. (2) uses the emission dipole pattern. For one factorized emission channel, the probability of remaining bound is the product of the three axis retentions; its complement is loss. Coherent paths also require cross-kernels. Resolved spherical emission channels omit their mutual interference; Cartesian marginals omit correlations within one photon's recoil. Scattering transfers populations but discards coherence transfer.

3 Calibration and numerical propagation

The carrier measurement fixes an intensity scale

The 5 kHz carrier measurement refers to $|2,2\rangle \leftrightarrow |3,3\rangle$ flops with only the 785 nm lattice present and pumping light off. In **modeled** decay mode it sets the root-mean-square local carrier frequency. Here j spans thermal states, positions and static Raman-lattice registration; the weights w_j sum to one. The 90% contrast is preparation/readout visibility.

$$f_{\mathrm{RMS}}^2 = \sum_j w_j f_j^2 = (5\,\mathrm{kHz})^2, \qquad P_{\mathrm{vis}}(t) = 0.9 \sum_j w_j \sin^2(\pi f_j t) \tag{3}$$

Equation (3) fixes the trace's initial quadratic rise and gives 1.363 mW per tone per beam (2.726 mW per beam). No homogeneous damping is fitted. The envelope $|\Sigma w_j \exp(2\pi i f_j t)|$ first reaches 1/e at 89 µs, or 0.445 carrier periods, compared with the reported 3-4 periods. A 700 µs exponential is shown only as a 3.5-period guide. The model does **not reproduce the reported decay**.

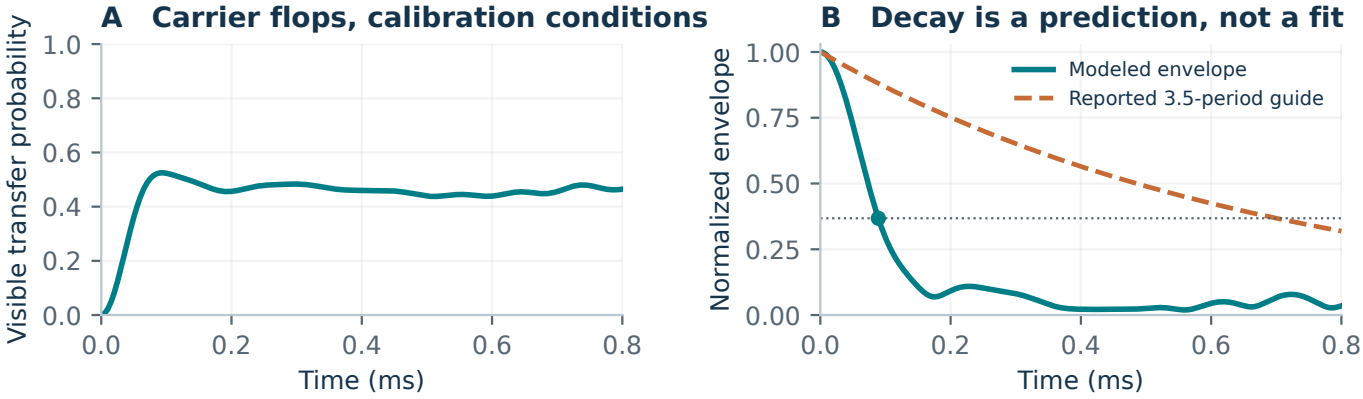


Figure 3. Cached carrier prediction under the calibration conditions. The dashed exponential is a comparison guide from the reported decay time, not measured trace data and not a fitted model component. The modeled envelope can revive.

A positive split-step evolution

Each constant pulse segment alternates a half-step Raman unitary, scattering and coherence damping, then a second half-step unitary (Strang splitting). The population generator Q is propagated by uniformization, a Poisson-weighted sum of nonnegative transition updates. Choose r at least as large as the maximum total event rate Γ_i; loss and photon counts follow the same population evolution.

$$e^{Q\Delta t} = e^{-r\Delta t} \sum_{k=0}^{\infty} \frac{(r\Delta t)^k}{k!} (I + Q/r)^k, \qquad r \geq \max_i \Gamma_i \tag{4}$$

An absorbing state completes conservation in Eq. (4) and is recorded as a loss counter. The reference uses a maximum 2 µs splitting step and 24-point recoil quadrature. The Poisson series is truncated to numerical precision; splitting error still requires convergence checks. Legacy ODE tolerances do not control this ensemble integrator.

A seeded, scrambled Sobol sequence supplies 64 equal-weight cloud-position/phase samples; calibration uses 128. **Frozen** means the cloud-envelope position and registration stay fixed during the 5 ms sequence; motion within each well evolves. Independent samples run in parallel (four workers by default, up to eight in the GUI). Cached responses and propagators reduce work; recorded optimization checks preserve the reference outputs bit-for-bit.

4 Reference predictions and observables

The reference 64-node run predicts the following outcomes. Absolute fractions use the **initially loaded, locally bound ensemble**; capture from the preceding trap is outside the model.

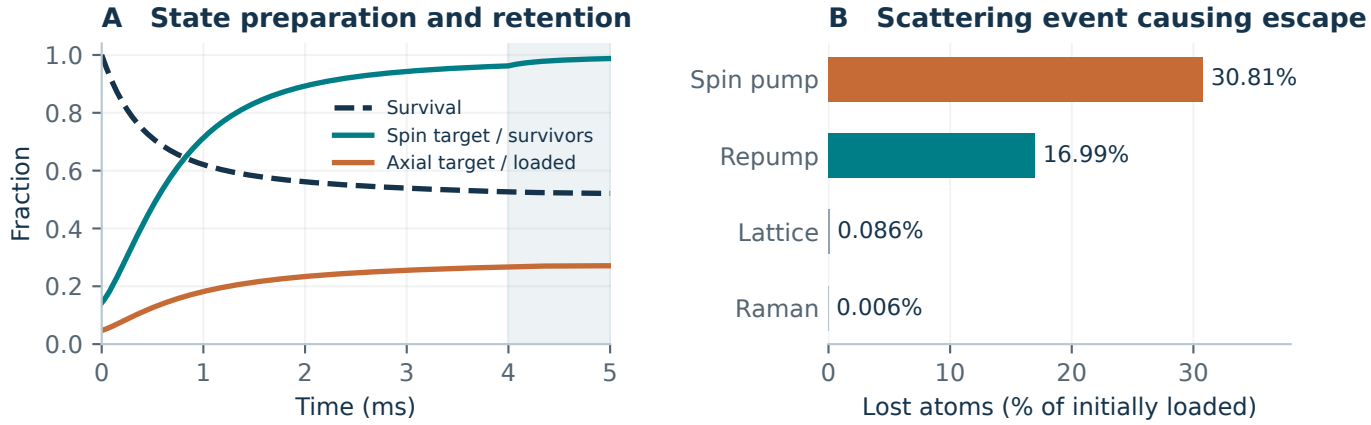


Figure 4. Left: survival, spin purity and absolute axial target yield; shading marks Raman off. Right: the scattering event causing escape, rather than each beam's indirect influence on the evolving state.

Observable at 5 ms	Initially loaded basis	Among survivors
Still bound in all axes	52.11%	-
Spin $\|3,3\rangle$	51.46%	98.77%
Axial ground state $n_z = 0$	27.51%	52.80%
Joint $\|3,3; n_z = 0\rangle$ target	27.09%	52.00%
Spin target and all three ground states	1.74%	3.34%

The denominator matters

Let S_j be a sample's total bound population and Y_j its bound population in $|3,3; n_z{=}0\rangle$, summed over transverse levels. Average these absolute populations before conditioning:

$$S = \sum_j w_j S_j, \qquad Y = \sum_j w_j Y_j, \qquad P(\text{target} \mid \text{survival}) = Y/S \tag{5}$$

Temperature proxies and loss selection

Axis energies are matched to finite-spectrum canonical mixtures weighted by local survivors, giving **6.53, 7.01 and 4.10 µK** for x, y and z. These proxies need not describe thermal equilibrium. Hot-atom loss can lower transverse temperatures without transverse Raman cooling.

Optical pumping dominates the modeled loss

Spin pumping and repumping account for 99.8% of the 47.89% well loss (Figure 4). Pumping events involving x/y escape produce 68.5% of all loss; 79.2% occurs in the first millisecond. Median local depths are about 10 µK, below the 15 µK central value. The loss-attribution rerun preserved all original population trajectories exactly.

5 Validation, limits and reproduction

Numerical evidence and physical uncertainty

The recorded suite passed 130 tests covering independent matrix exponentials, recoil and polarization conventions, conservation and parallel execution. The canonical run conserves survival plus loss to 3.7×10^{-14}. Numerical consistency does not establish the accuracy of the physical approximations.

Check	Recorded change / interpretation
Timestep 2 → 0.5 µs	Axial target change -5.90×10^{-7}, earlier matched four-node reference
Recoil quadrature 24 → 48	Target change below 10^{-14}, same earlier reference
Spatial nodes 32 → 64	Current target shifts +0.705 percentage points; residual dependence remains
Optical reduction	Excited fraction 6.17×10^{-4}; pumping rate / smallest local motional angular spacing reaches 0.480 (marginal)

The timestep and recoil checks used an earlier calibration and z-lattice polarization. The secular approximation is best justified when pumping is much slower than local motional oscillation; a maximum ratio of 0.480 is marginal despite weak excited-state occupation. No statistical confidence interval or total physical error bound is assigned.

The principal model boundaries

Confinement and loss: isolated separable scalar wells; no tunneling, intersite transport, recapture, collisions or coherent Raman escape into the continuum. Leaving one well is counted as loss even if an apparatus might recapture the atom. **Optical dynamics:** population-resolving scattering omits coherence transfer and full emission correlations. Static light shifts are diagonal; higher-state polarizability corrections and state-resolved excited-state lattice shifts are absent. **Calibration:** spatial registration is frozen, technical phase noise is unspecified, and the reported carrier decay is not reproduced. These limits matter especially when increasing pumping power.

Reproduce the reference

From the repository root, use the **scripts** conda environment. To run the canonical file with eight workers without changing its saved settings:

```
conda run --no-capture-output -n scripts python rb85_rsc_sim.py
  --config examples/experiment.json --headless
  --set ensemble.workers=8 --output results/whitepaper_check
```

Enter the command as one line. Reference snapshot: commit **5558241**; configuration SHA-256 begins **f13e18574eef1519**. Sources are results/astra_modeled_decay/final, the modeled carrier cache and the matched loss diagnostic. This PDF embeds **reference-data.json** with the input configuration, plotted data and provenance.

References and implementation record

[1] D. A. Steck. Rubidium 85 D Line Data, rev. 2.3.4 (2025). steck.us/alkalidata/rubidium85numbers.pdf

[2] N. Šibalić et al. ARC: An open-source library for calculating properties of alkali Rydberg atoms. Comput. Phys. Commun. 220, 319-331 (2017). doi:10.1016/j.cpc.2017.06.015

[3] F. Reiter and A. S. Sørensen. Effective operator formalism for open quantum systems. Phys. Rev. A 85, 032111 (2012). doi:10.1103/PhysRevA.85.032111

[4] R. Ozeri et al. Errors in trapped-ion quantum gates due to spontaneous photon scattering. Phys. Rev. A 75, 042329 (2007). doi:10.1103/PhysRevA.75.042329. Cited for scattering mechanisms, not Rb-85 loss validation.

Implementation: rb85rsc/{ensemble,far_detuned,motion,optical,model}.py. Detailed assumptions and recorded checks: docs/astra-physics-corrections.md; execution and benchmarks: docs/experiment-workflow.md and docs/performance.md.